



World's Largest Stevia Project — NF & RO System

Concentration of Stevia Glycosides by NF & Process Water Recovery by RO · Peru, South America

SITE INFORMATION

LOCATION

Peru, South America

INDUSTRY

Herbal

SYSTEM SUPPLIED

NF & RO

NF-I CAPACITY

70,000 litres/hr

NF-II CAPACITY

30,000 litres/hr

RO CAPACITY

90 KL/hr

- >99% Stevia Glycoside retention by NF
- >95% product recovery achieved
- NF-I: ~20x concentration (>95% dewatering)
- NF-II: concentrates to 15% TS before evaporation & spray drying
- RO permeate recycled as process water for resin columns
- FDA sanitary design; fully automatic

OBJECTIVE

Concentration of Stevia Glycosides by NF and process water recovery by RO for resin column dilution.

SCOPE

Fully Automatic NF I, NF II & RO systems for Process and CIP cycles. High-end, completely automatic FDA sanitary design installed in the world's largest Stevia plant.

TECHNOLOGIES

- Nanofiltration — Stage I (NF-I)
- Nanofiltration — Stage II (NF-II)
- Reverse Osmosis (RO)
- Fully automated PLC, FDA sanitary design

>99% Stevia Glycoside Retention · >95% Product Recovery · NF-I: 70,000 L/hr · NF-II: 30,000 L/hr · RO: 90 KL/hr

INTRODUCTION

Permionics supplied the Nanofiltration and Reverse Osmosis systems for the world's largest Stevia plant, located in Peru, South America. The installation is a fully automatic, FDA sanitary design operating at a combined NF capacity of 100,000 litres/hr, making it one of the largest membrane-based natural extract concentration systems deployed globally.

The process handles the concentration of Stevia Glycosides from a dilute aqueous extract through two sequential NF stages, with a companion RO system recovering and recycling process water from the NF permeates back into the resin column dilution steps.

THE CHALLENGE

Feed Characteristics & Process Demand. The aqueous extract of Stevia post clarification is typically composed of approximately 4% TDS and has approximately 35% purity. The downstream process requires this to be purified to approximately 95% and concentrated to approximately 15% Total Solids (TS) before evaporation and spray drying — a demanding multi-step separation requiring precise membrane selection and staging.

Stage	Feed Condition	Requirement / Target
Feed to NF-I	~4% TDS, ~35% purity	Concentrate ~20x (>95% dewatering) before Resin Step I
After Resin Step I	Purity improved ~35% → ~75%; diluted to 0.3–0.4% concentration	Concentrate via NF-I; remove small MW impurities & salts
After Resin Step II	Purity ~95%; diluted ~10x to ~0.8% concentration	Concentrate via NF-II to 15% TS before evaporation & spray drying
NF Permeates (I & II)	Contains minor ash and organics	Polish via RO to obtain process water for resin column dilution

THE TECHNICAL SOLUTION — PROCESS FLOW

NF Stage I (70,000 litres/hr)

- ✓ Purification of extract by resin in the first step improves purity from approximately 35% to approximately 75%, but this step dilutes the extract — the output of Resin Step I has a concentration of only 0.3–0.4%. Permionics NF-I (70,000 litres/hr) concentrates this approximately 20 fold (>95% dewatering) before it goes to Resin Step II. The NF membrane also facilitates removal of small molecular weight impurities and salts.

NF Stage II (30,000 litres/hr)

- ✓ Resin Step II further purifies the product from 75% to >95%; here again there is a dilution by 10-fold, resulting in a ~95% pure stream at 0.8% concentration. This is further concentrated by NF-II (30,000 litres/hr) to 15% TS before evaporation and spray drying.

RO System (90 KL/hr)

- ✓ Permeates of both NF-I and NF-II are further polished in the RO system to remove minor ash and organics, yielding process water that is recycled back for resin column dilution.



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PERFORMANCE RESULTS

- Commercial scale data at Peru site is consistent and proved that integrating NF after both Resin steps retains >99% of Stevia Glycosides
- NF plays an important part in reducing load on resin columns by removing smaller molecules and ash
- >95% product recovery achieved
- Integrating RO provides process water for recycle, closing the loop

DESIGN STANDARD

High-end, completely automatic FDA sanitary design. Installed in the world's largest Stevia plant. Fully automatic NF I, NF II & RO systems for both process and CIP cycles.

VALUE DELIVERED

- Reduces load on resin columns — removing smaller molecules and ash before each resin step
- Concentrates the diluted resin-step output back to processable levels at both stages
- Recovers process water via RO permeate for reuse — zero process water waste
- >99% Stevia Glycoside retention throughout the NF stages

FOR MORE INFORMATION

Permionics Membranes Pvt. Ltd.

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Process Solutions Division
India · Global

OPERATIONAL RESULTS

>99% STEVIA GLYCOSIDE RETENTION BY NF	>95% PRODUCT RECOVERY ACHIEVED	~20x NF-I CONCENTRATION FACTOR	15% TS FINAL CONCENTRATE BEFORE EVAPORATION
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Performance Parameter	System	Result
Stevia Glycoside retention	NF (both stages)	>99% retained; >95% product recovery
Concentration factor — NF-I	NF-I (70,000 L/hr)	~20 fold (>95% dewatering) of resin step I output
Feed to NF-II purity	After Resin Step II	~95% purity at 0.8% concentration
Final concentrate — NF-II output	NF-II (30,000 L/hr)	15% TS — ready for evaporation & spray drying
Resin column load reduction	Both NF stages	Smaller molecules and ash removed before each resin step
RO permeate quality	RO (90 KL/hr)	Process water recovered for resin column dilution — recycled
Design standard	Full system	Fully automatic FDA sanitary design

INSTALLATION PHOTOS



Peru Stevia Project Installation — Top View (June 2018). NF and RO membrane skids spanning the full production floor.



Peru Stevia Project Installation — Side View (June 2018). NF pressure vessel arrays with process tanks in the background.

CONCLUSIONS

The Peru Stevia NF-RO project stands as proof of Permionics' capability to design and deliver large-scale, fully automatic membrane systems for complex natural extract concentration at the world's largest level. Commercial scale data at the Peru site is consistent and has proved that integrating NF after both resin steps retains more than 99% of Stevia Glycosides, plays an important role in reducing load on the resin columns by removing smaller molecules and ash, and delivers greater than 95% product recovery. The integrated RO system further closes the loop by recovering process water from NF permeates for recycle into the resin column dilution steps — eliminating process water waste while maintaining the stringent FDA sanitary design standards required for herbal pharmaceutical production.